Compiler repeated work: profile, reduce, verify

Created-at: 2026-09-07 12:48 +02:00

Scope and ETA. Investigate cold source compilation, AST copying, reference resolution and repeated traversals. Approximately 45 minutes: 10 minutes instrumentation, 10 minutes profiles and baseline, 15 minutes implementation and correctness, 10 minutes repeat measurements and report. Generation, parsing, lowering, hashing, prepared compilation and destruction remain distinct measurement scopes. Do not remove validation or change hashes.

Authority and checkout. User requested optimization and investigation of repeated calls. Core fetched successfully at 12:42–12:45 using the existing origin with an SSH HostName override for 192.168.1.228; normal DNS failed. Base is freshly fetched origin/main f305ade; branch perf/compiler-repeated-work reuses the existing checkout. Former branch 2d43e82 has three unmerged diagnostic commits and is preserved. Untracked distributed-verification artifacts are preserved.

Discovery. Extend the existing interning Criterion benchmark and ModuleStore integration tests. ir_bench explicitly excludes parsing/lowering; ir_depth_probe measures depth failure, not compilation. Historical CPU optimization and CBOR reports contain no equivalent compiler profile. Ecosystem reader confirms Core owns compilation and is consumed by Graph, Transformation and Python.

Affected files. This four-format plan; `hymeko_core/benches/interning_bench.rs`; `hymeko_core/tests/mir` candidate production files `hymeko_core/src/module_store/module_store.rs`, `hymeko_core/src/ir/lower.rs` `hymeko_core/src/ir/canonical_hash.rs`, `hymeko_core/src/resolution/resolve.rs`; `reports/2026-09-07` and associated numerical, profile and figure evidence under `reports/compiler-repeated-work/`.

Core protection. User approved at the 2026-09-07 continuation: `APPROVED-CORE-EDIT: compiler-repeated` Instrumentation is allowlisted; candidate production files are lockdown-full. Apply the prepared root-AST/dependency-list borrow patch first, keeping imports, hashing and validation unchanged. Compare exact IR output (only timestamp normalized), then cold compilation and cached lookup. Preserve before/after binaries and PDBs. The user's runtime question also requires quoting the existing prepared Dijkstra results separately: compiler speedups are not traversal-kernel speedups.

Interfaces and invariants. Retain public signatures, cache behavior, resolution ambiguity/error order, import namespace order, IR contents and hash bytes. The legacy Merkle pass writes serialized decl_hash and is not redundant merely because another hash pass exists. No dependency changes. If borrowing ASTs requires an API change, keep the public owned-slice API and share the underlying lowering.

Measurement and tests. Criterion release, minimum five post-warmup iterations; save raw samples, median/IQR/worst. Windows samply CPU profile with symbols before selecting a hot spot. Read existing Graph campaign artifacts first. One large production-scale smoke precedes

repeated large runs. Record process peak working set, host and toolchain. Differential whole-IR checks normalize only metadata timestamps and pin both canonical and declaration hashes; import, constants, aliases, empty inputs and failure cases. Run relevant and workspace tests, fmt, clippy, symbols, duplication and preflight. Benchmark same machine and build settings.

Budgets and risk. Maximum 16 GiB RSS, 180 seconds per measured process; the existing 50,000-node source provides a quick discriminating workload, followed by the graph map512 source (30,032,345 bytes) if available through the existing harness. Target measurable reductions without greater than 10 percent time or memory regression. No claimed speedup without profile attribution. Nested imports and ambiguous reference paths can invalidate naive early return or memoization; tests must preserve them. Cloning may contribute memory rather than dominant CPU cost. Retaining cache entries may exchange speed for unbounded memory, so do not extend cache lifetime speculatively.

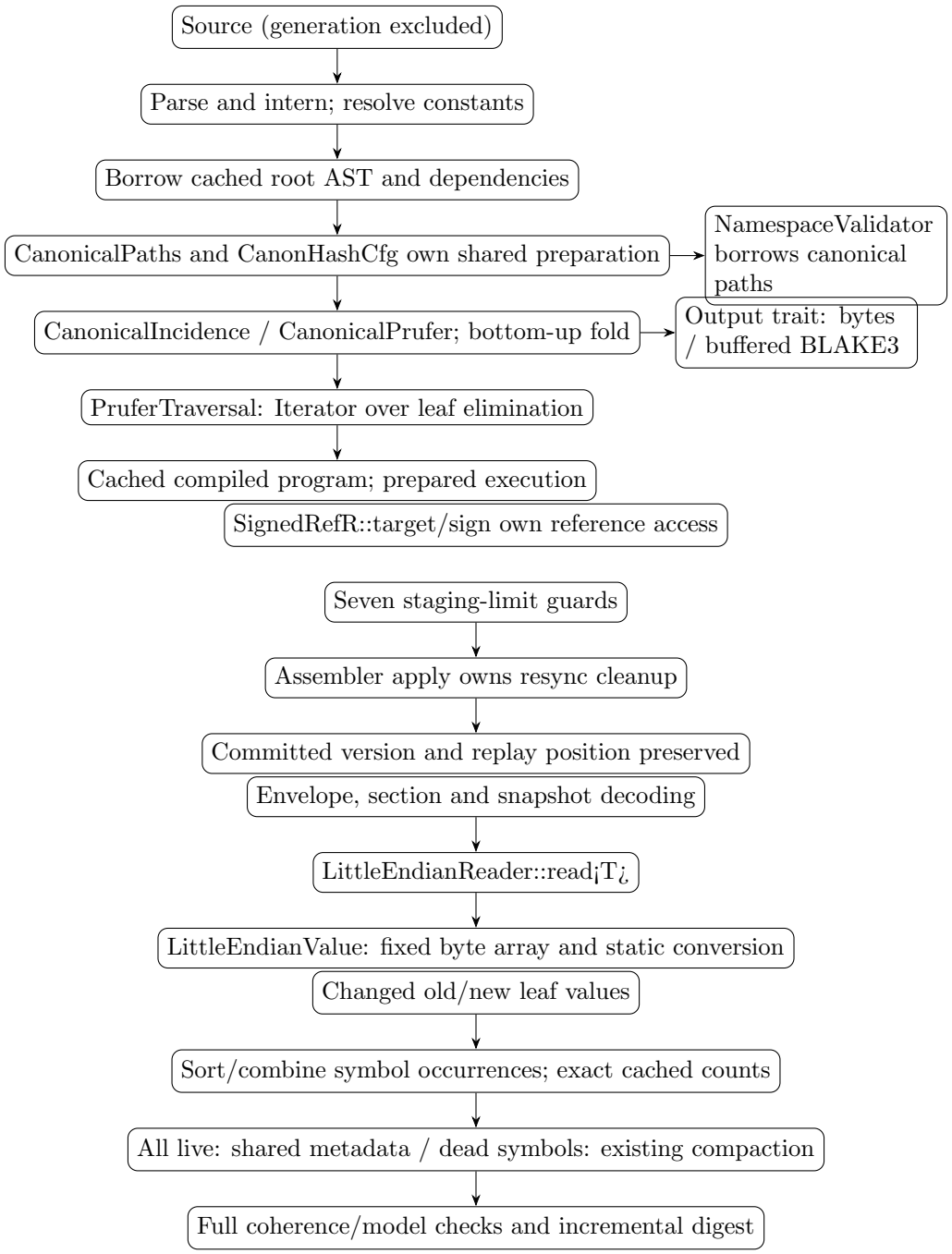
Rollback. Revert only this task's named changes. Preserve user files, previous branches, benchmark data and public hash/wire contracts.

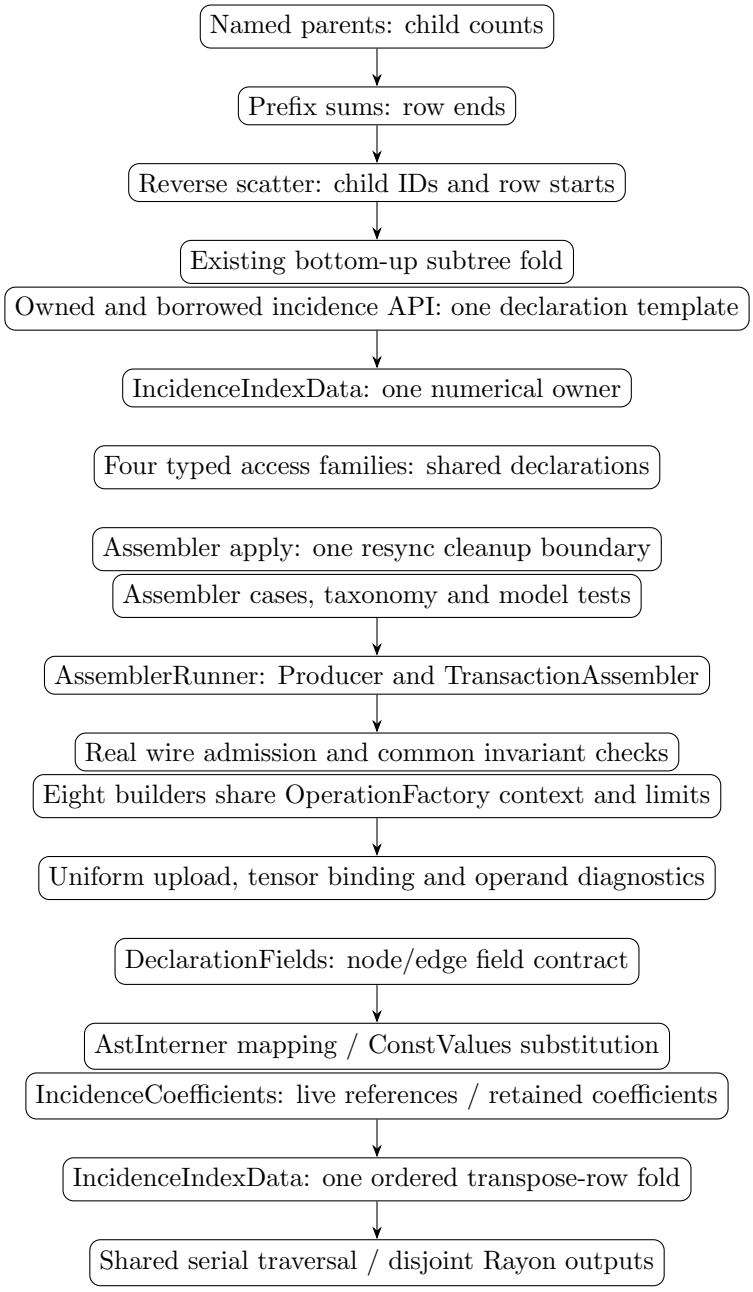
Continuation, 2026-09-07 13:36. Under the same approved slug, strengthen source-to-IR compilation by reusing the canonical path table between validation and hashing. A crate-private CanonicalPaths view borrows the IR and owns just its derived paths; the IR cannot change during its lifetime. Public standalone APIs delegate to the same methods. Validation borrows index paths rather than cloning them, preserving the original index iteration and first-error order. Existing DerivedPathIndex also sorts a lookup table and is therefore inappropriate for this transient compile phase. Extend existing ModuleStore tests to compare compiled and standalone hashes, and to exercise namespace corruption. Frozen hashes and full serialized-IR goldens remain unchanged. Compare the already optimized root-borrow binary against this one change first; only then consider repeated leaf hashing. Expected additional work: 20 minutes including two bounded measurement pairs, tests, coverage and report. Retain the 50k baseline and add a larger existing frontend-generator scale if the smoke fits the declared 16 GiB / 180-second process budget. No second benchmark framework.

Continuation, 2026-09-07 13:59. Profile-guided compact Pruefer state: the resolved final profile puts 19.1 percent of compile-stack samples in prufer_from_forest. Replace its HashMap of dense declaration IDs with a dense vector and per-vertex adjacency lists with degree plus neighbor XOR. For degree one, the XOR equals the sole remaining neighbor, and removing an edge XORs out its endpoint. Preserve the existing heap key/order, labels, token sequence, super-root handling and hash domains. No public API or dependency change; same Core approval. Extend hymeko_core/tests/unified_ir/verification/frozen_hash.rs before production edits with deterministic chains, stars, balanced trees, multiple roots, empty and single-node forests, interleaved anonymous declarations and nondefault hash configuration. Freeze an aggregate digest against the current shared-path version before the rewrite. Reuse its archived binary as the immediate baseline. Measure cold compile and standalone canonical hashing at 50k, then cold compile at the already validated 500k scale in ABBA order; caps remain 180 seconds and 16 GiB per process. Capture a resolved after-profile, execute workspace tests, clippy and coverage, and append to the same report. ETA about 15 minutes: targeted parity/build 3, bounded A/B 6, profile plus checks/report 6. Roll back only this continuation if parity fails or investigation finds a regression. Existing root-borrow and shared-path edits remain.

Continuation, 2026-09-07 20:06. Remove duplicate annotation-value resolution in resolve_signed_refs under the existing `APPROVED-CORE-EDIT: compiler-repeated-work`. Fresh fetch confirms HEAD equals origin/main f305ade; preserve all earlier dirty changes. The exact current profile attributes

610 of 8070 compile samples to this function. Resolve the annotation once and clone its resolved value into the existing weights field; both owned wire fields must remain. Preserve list depth, absent versus empty, floating-point bits, symbol insertion order and first error. Extend existing test_ref_values.rs with mixed-value and error controls before editing production; extend the normalized-IR corpus for the real-parser path. Extend the existing Criterion generator with an explicit mixed-reference payload, retaining the numeric fixture and distinct benchmark identities. Archive matched binaries; compare lowering and cold compile in ABBA order, ten samples after warmup, with raw samples, IQR, worst and process peak. Existing 50k and 500k numeric fixture bounds apply; mixed 50k smoke must precede repeated runs. Each process capped at 180 seconds / 16 GiB. ETA approximately 20 minutes: tests and baseline 5, implementation and measured pairs 7, workspace gates and report/audit 8. No API or dependency changes. Bounded audit covers ModuleStore, intern/constant/reference resolution and lowering; record proven repeated work separately from unmeasured candidates and necessary semantic passes. Other candidates are not automatically bundled into this patch.





PERF-1 integration, 2026-09-09

Created-at: 2026-09-09T18:54:44+02:00. ETA: approximately 60–90 minutes, based on cached builds, existing 1-minute Core tests, 1–3 minute paired benchmark cells, SDK rebuild and browser tests; no complete historical campaign rerun. User approved saving the four-step strategy and executing step 1 in this turn. Existing authorization: `APPROVED-CORE-EDIT: compiler-repeated-work`; the course-module-compilation approval was the user's affirmative reply to that named scope, not a literal token. No new semantic or accelerator work.

Reuse the released course branches (Parser a528d98, Core 8623eb1) as bases. The original dirty performance checkouts remain the preserved source of the measured edits; independent integration worktrees hold the released control and candidate during A/B builds. Core main's independent sparse commit is not silently merged. Retire the temporary worktrees after evidence and publication.

Integrate only the existing grammar/string-scan optimizations, owned-record benchmark adapter, root borrowing/shared canonical paths, compact Pruefer state and single annotation-value resolu-

tion, plus their existing tests. Keep final course tests and diagnostics from the published branches. Affected additional paths: Parser grammar/lexer, existing benches/formats, tests/format_comparison.rs, scripts/format_*.py and their tests, Parser Cargo files and existing plan; Core canonical_hash.rs, module_store.rs, resolve.rs, existing interning benchmark and frozen/ref/module-store tests, Cargo pin/lock, generated SYMBOLS.md and docs/hymeko-platform-strategy.md; SDK Cargo pin/lock, existing changelog/package README and generated package. Root report exclusions remain. No Graph/Transformation/Tensor source change.

Use identical harnesses to build released control and integrated candidate. Run large-case smoke before bounded measurements: 50k linked/long-text parser and 50k mixed Core; 500k Core correctness/resource smoke where affordable. Minimum five post-warmup samples, retain raw values, median/IQR/worst and peak memory, binary/source/fixture hashes, affinity and environment. Reuse existing Campaign/ProcessRunner, Criterion and plotting tools. A/A or repeated control bounds ambient noise. More than ten percent stable regression requires investigation; inconclusive is not pass. Existing profiles justify the already-measured optimization mechanisms, not a fabricated combined speedup. No tests/builds overlap timed cells. Max process peak 16 GiB, timed process 180 seconds. Preserve full-IR and corpus hashes, signed-reference diagnostics, source provenance; run Parser, Core, native/Node/browser/Worker parity and actual packed SDK installation. CPU tests only; no GPU performance claim.

Publication: provider before consumer, immutable Git revisions, one Core identity, no local path patches shipped; verify GitHub and LAN refs. Reuse current course branch without default-branch merge. Archive reports to Samba. Rollback uses previous published SHAs and retained original SDK; never reset or remove unrelated user work. PERF-2/3/4 remain planned at task completion.

Current full-IR characterization, 2026-09-13

Created-at: 2026-09-13 15:37 +02:00. ETA: about 15 minutes for existing target builds, correctness, bounded cells and report. User explicitly requests the full IR benchmark and the compiled library size. No circular-buffer remeasurement, profiler, production change or second compilation pipeline. Reuse ModuleStore::compile through the existing interning benchmark: 50k numeric and mixed sources, cold compile and cache hit reported separately. Reuse existing canonical-CBOR corpus encode/decode cells as independent boundary measurements; do not add phase medians to claim an end-to-end result. Ten or more post-warmup Criterion samples, median/IQR/worst of per-sample iteration averages, no per-call p99 claim. Run the exact public-facade corpus round-trip correctness test first. Use existing measure_peak_memory.ps1 Program mode with explicitly built targets, at most 16 GiB observed working set and 180 seconds per process. A single same-scale smoke precedes repeated cells. Build/check work never overlaps timing. Fresh release library build: default-empty features, no IPC/Arrow/GPU; report rlib versus cdylib/PDB and inspect DLL exports. A small DLL without callable Core exports is not the full linked Rust dependency footprint. Affected tracked files: this existing plan.tex and rebuilt plan.pdf only. Append results to reports/2026-09-07-compiler-repeated-work.md and reuse its compiler-repeated-work evidence directory for raw samples, resource records and plots. Preserve the other session's uncommitted circular-buffer bench, manifest and plan changes. Rollback: revert this plan addendum; no runtime migration.

Post-extraction audit and performance, 2026-09-23

Created-at: 2026-09-23 23:10 +02:00. ETA: 10–15 minutes including release builds. User explicitly requested duplication inspection followed by performance benchmarks. Reuse current dirty Core, Tensor and Nagare checkouts; no commits, worktrees, source backups or production optimizations. Fetch succeeded in all three. Pin the exact source hashes in the measurement

record.

Audit the combined production sources of Core, WGPU, Tensor's numeric crate and Nagare HNN with the existing token detector, using the same scope for Git HEAD and current files. Report relocation separately from removed duplicates. Inspect the surviving clone groups and distinguish shared contract duplication from CPU/GPU strategies and different mathematical operators.

Artifact ledger: extend the existing interning benchmark with a semantic-digest characterization group (full digest, cached read, checked leaf update including clone/validation); reuse its deterministic source generator and public Core API. Verify equality against the full digest before timing. Reuse the existing CSR benchmark without copying its kernels or fixtures. Keep measurement opt-in. Write reports/2026-09-23-core-tensor-audit-performance.md and raw JSON/logs/plots under the existing reports/compiler-repeated-work directory, prefix cutover-. No new runner or plan directory. Existing peak-memory runner and Criterion own execution and sampling; no build/test overlaps measured cells.

Measure 50k-node cold/cached compilation and small/medium CSR SpMV/SpMM at 24 Rayon threads. Semantic digest uses a bounded 1000-node fixture with one editable value. Minimum 10 warmed-up samples; median, inclusive IQR and worst sample average (not per-call tail latency). Use historical same-host samples only where source/config match; absence is a new baseline, not evidence of speedup. Report OS, CPU, Rust version, flags, fixture digest and process peak working set. Userspace RSS guard: 16 GiB; stop an individual timed process at 180 seconds. A greater than 10 percent change is investigated before a causal claim. Retain raw results even if the host is noisy. CPU results only; no GPU/remote claims. Archive the report using the existing Samba synchronization tool. Rollback of benchmark-only changes uses Git; generated measurement evidence stays in reports.

Canonical incidence owner, 2026-09-24

Created-at: 2026-09-24 00:23 +02:00. ETA: 8–12 minutes for the focused refactor, cached targeted tests and report. User explicitly requested a struct/trait-oriented incidence API with hash() and bytes(); this authorizes the Core canonical hashing edit. No benchmark or dependency change.

Extend canonical_hash.rs with CanonicalIncidence, borrowing immutable IR, and a private output trait implemented by Vec¡u8¿ and the existing BLAKE3 hasher. One record iterator and encoder serve both methods. Normal hashing streams ordered edges. Duplicate paths or wrapped u16 positions require sorting lightweight records, using the same encoder without a byte arena or fallback pipeline. Preserve the existing byte order, domains and hash config. The existing public program-hash facade delegates to the owner; no old incidence free functions or forwarding aliases remain.

Affected files: canonical_hash.rs, existing frozen_hash.rs tests, generated SYMBOLS.md, this four-format plan and the existing Core/Tensor audit report. Fresh fetch of all Core remotes succeeded; HEAD 10138bf equals the task upstream and is three commits ahead of origin/main. Preserve all prior dirty work; no new checkout, backup, commit or publication.

Verification: fixed payload bytes and header-plus-payload digest equality; empty incidence, mixed signs, repeated targets, edge ordering, duplicate edge paths, u16 arc/slot wrapping and long path truncation. Existing frozen whole program vectors and generated forest/config corpus remain unchanged. Run only the affected correctness target, library lint and touched-file format checks. Worst input has many incidences with colliding paths: sorting holds record descriptors, never duplicate encoded payloads. No speed or memory improvement claim without measurement. Rollback is the scoped Git diff; no source backup. Archive the amended report through the existing shared report-sync tool.

Pruefer owner, 2026-09-24

Created-at: 2026-09-24 00:47 +02:00. ETA: 8–12 minutes for a focused source refactor, cached correctness checks and report. The user explicitly requests struct/trait orientation for Pruefer, authorizing the Core edit. Core fetch succeeded on origin and github; task HEAD/upstream remains 10138bf, three commits ahead of origin/main. Preserve existing uncommitted work.

Replace the tuple forest and free Pruefer orchestration with SkeletonForest methods, a prepared CanonicalPrufer owner, and PruferTraversal implementing Iterator. Store degree, neighbor XOR, live vertices and leaf heap in the traversal. Share the existing CanonicalSink trait for byte/hash output; one token iterator determines both. The canonical program hash uses this owner directly. No compatibility wrappers or duplicate algorithm remain. Preserve leaf priority (subtree hash, path tie, vertex), super-root handling, iterative subtree folding, token order, headers and malformed-input behavior.

Affected files: canonical_hash.rs, existing frozen_hash.rs tests, generated SYMBOLS.md, this four-format plan and the current Core/Tensor audit report. Update stale helper-name comments in coherence.rs and the existing hierarchy and test-oracle files; these edits do not change executable behavior. Extend fixed hash vectors with public byte/hash parity, empty/single/multiple roots, interleaved anonymous declarations and generated forest shapes/configs. Exercise invalid cycle behavior explicitly. Run the frozen-hash and existing ordinary hierarchy correctness cases, excluding the ignored resource probe; library Clippy and touched-file formatting. No benchmark or dependency change. Worst-case deep trees remain heap-traversed; hashing streams fixed-size tokens, without constructing the full Pruefer payload. No measured speedup is claimed. Rollback is the scoped Git diff, preserving the preceding incidence change. No new checkout, source backup, commit or publication. Archive the final report.

Remaining canonical helpers, 2026-09-24

Created-at: 2026-09-24 00:57 +02:00. ETA: 6–10 minutes for the Core boundary; a public API migration may additionally require downstream checks. User requests remaining free functions to move into their responsible structs. Core fetched both remotes successfully; existing HEAD/upstream 10138bf remains.

CanonHashCfg owns domain-header initialization and chunk hashing. CanonicalPaths owns construction from IR, reverse paths for resolution-boundary validation and resolved strings. SkeletonForest owns its node-domain mapping. Reuse these owners from Pruefer, incidence, path-index construction and program hashing; remove their free helper bodies. No new struct solely to hold static utilities. The user explicitly approved public entrypoint removal with consumer migration. Promote CanonicalPaths::new/hash/validate; add DerivedPathIndex::new. Delete the three free public entrypoints and migrate Core, web, XR, Kawamirai, Transformation, Graph, the C++ Rust oracle and Nagare call sites. Keep manifest and lockfile pins unchanged; local verification uses command-line path patches. DerivedPathIndex is the existing owner for path-index construction. The XR check also reaches the existing simulator-xr-publish checkout. Migrate its callers and the primary simulator, Python binding and robot middleware found by the workspace-wide search. Exclude inactive alternative checkouts and the retired monorepo. Python's section-name adapter additionally requires the already-existing CanonicalIrCbor enum case to compile against current Core.

Discovery ledger: find canonical_hash.rs and existing consumer Rust files; grep the three public names and all private helpers across framework/Nagare; symbols confirm CanonHashCfg, CanonicalPaths, SkeletonForest and DerivedPathIndex already own these concerns; read implementations and callers. Verdict: EXTEND existing owners, remove free bodies, migrate callers in place.

Affected Core files: canonical_hash.rs and focused frozen/path/namespace tests; the existing plan bundle, symbol index and audit report. If public entrypoints are migrated, their Core

consumers and downstream callers are also in scope. Keep canonical bytes, error variants and iteration behavior unchanged. Verify fixed hash vectors, byte/hash parity, namespace failures, duplicate paths and normal lookup behavior. No benchmark or dependency upgrade. Preserve prior edits; rollback is only this continuation's diff. No new checkout or source backup.

Namespace validator owner, 2026-09-24

Created-at: 2026-09-24 01:15 +02:00. ETA: 3–5 minutes. User requests a dedicated validator struct. Core fetch succeeded on both remotes; task HEAD 10138bf equals upstream and remains three commits ahead of main. Preserve the existing checkout and all preceding changes.

Discovery ledger: search IR/resolution for Validator and Validation owners finds none; CanonicalPaths in SYMBOLS.md and its implementation own the present validation. Read the method, namespace errors and all validation call sites. Verdict: EXTEND canonical_hash.rs with NamespaceValidator borrowing an existing CanonicalPaths, and MOVE validation plus reverse-index preparation into it. Remove CanonicalPaths::validate, with no forwarding method or free wrapper. Keep namespace errors and canonical hash bytes unchanged. Migrate lowering, CBOR validation and existing tests/bench call sites; retain shared path preparation. Run namespace/frozen-vector and lowering-boundary correctness tests, CBOR tests, compile the affected bench and run library lint. Update symbols and duplication gate, append the existing report and archive. No measurements or dependency edits. Rollback is the scoped diff; no backup or new checkout.

Subtree fold optimization, 2026-09-24

Created-at: 2026-09-24 01:21 +02:00. ETA: 4–6 minutes. User requests a more efficient subtree-hash implementation. Both Core remotes fetched; HEAD 10138bf still equals task upstream. Preserve current checkout and prior edits.

Discovery: SkeletonForest::new is the sole owner of parent/children/root construction; subtree_hashes is its sole fold. Each named declaration is inserted once under at most one valid named parent, or into roots. Therefore a root-reachable cycle or shared child is impossible even for malformed input; cycles and their descendants are disconnected from roots and retain zero hashes. Verdict: EXTEND the existing method, no additional module or production owner.

Build one root-first order, fold it backwards, and write directly into a dense zero-initialized HashId vector. Remove Enter/Exit frames, per-node Option state, on-stack flags and final conversion. Reuse one child-hash scratch buffer sized to maximum reachable fanout; skip it for leaves and single-child nodes. Use in-place unstable lexicographic sort: equal hashes contain identical bytes, so stability has no semantic effect. Cache the two configured named-leaf hashes within the fold, never across calls or configurations. Preserve all hash bytes and disconnected-component behavior. Complexity remains linear traversal plus sum of child-sort costs; no measured speedup is claimed.

Verify existing frozen vectors/configuration corpus, dense-ID permutations, deep/wide trees, mixed node/edge leaves, incidence multiplicity and malformed cycles. Extend the existing cycle regression to include a rooted component and a descendant of the disconnected cycle. Run only these correctness tests and library Clippy; no benchmark or profile was explicitly requested. Update the existing report, diagrams and symbol/duplication gates, then archive evidence. Rollback is the scoped diff; no backup, new checkout or dependency change.

Current Core performance, 2026-09-24

Created-at: 2026-09-24 01:30 +02:00. ETA: 3–5 minutes. The user explicitly asks about performance benchmarks. Measure the existing interning Criterion harness only: 50k canonical hash, cold/cached compile, and 1k full/cached digest plus checked leaf update. Reuse the existing

memory runner and raw-sample summary machinery; no new benchmark implementation, source backup or worktree. Both Core remotes fetched, HEAD 10138bf equals the task upstream.

Discovery/cached evidence: the September 23 report has paired compiler/digest results. Its executable still exists and SHA256 matches 84f5f2261a8db3946b3d- afa17fd789ba589c038006d40f0ce7ec1d16c5de Run it twice before rebuilding, then the current source twice, with explicit –bench, numeric 50k fixture, 24 Rayon threads, 1-second warmup and existing 10/20-sample groups. This compares the accumulated canonical refactors, not the isolated last subtree change. Verify source and canonical fixture hashes across binaries. Build time is excluded.

Budgets: each measured process below 16 GiB observed peak working set and 60 seconds wall time, expectation approximately 30 seconds from the prior 24.8-second five-cell run plus the hash cell. Preserve raw samples, report median/IQR/worst sample average and two-run drift. Memory polling uses the existing 100ms userspace cap; record its limits. Run sequentially without builds or tests during timing. The Windows/275HX interactive host has background GUI activity (largest observed process 0.156 CPU seconds per second); do not claim a quiet-host release gate or attribute every difference to the subtree fold. Investigate differences exceeding 10 percent within the measurement scope; profiling remains unrequested, and causal speedup claims require separate evidence. Keep source/manifest/executable provenance, amend this report and archive results. No changes to production source, dependencies, public APIs or published pins.

Canonical hash batching, 2026-09-24

Created-at: 2026-09-24 01:45 +02:00. ETA: 6–10 minutes including archive. Continuation of the requested Core performance measurements and optimization. Owner: Core canonical hash; the existing user authorization covers this private implementation. Both remotes fetched; HEAD equals task upstream, three commits ahead of main. Preserve unrelated dirty work and all public contracts.

Discovery ledger: CanonicalSink is the sole output strategy in canonical_hash.rs; no buffered hash sink exists in Core. Extend that strategy with one private CanonicalHasher (configuration/header, fixed 16 KiB buffer, write, finish). Replace the direct Hasher sink, not the encoder. Existing frozen_hash tests, interning_bench, memory runner, report and plan bundle are reused. New measurement artifacts belong to reports/compiler-repeated-work with batching prefixes.

Evidence: the existing resolved Pruefer profile includes 463 Hasher::update samples, 255 ChunkState::update samples, and 371 path-string samples. These are historical inclusive observations, not current causal attribution. Current source issues per-field 1–32-byte updates. Test the bounded batching hypothesis against the just-measured current binary (canonical hash 78.743 ms); do not claim it explains the historical 6.6 percent regression without isolation. Batch the 24-byte header in CanonHashCfg and pass the existing sorted child-hash array as one contiguous slice. No alternate pipeline, dependencies or hash-version change. Fixed buffer memory is O(1), independent of model size.

Acceptance: identical frozen hashes, configured headers, long UTF-8 prefixes, wrapped incidence slots, empty streams, exact/full/partial buffer boundaries, and repeated calls. Existing targeted correctness gate and library Clippy. Two before and two after runs of only 50k canonical hash and cold compilation, existing warmup/sample counts, sequentially, no concurrent build or test. Expected 12–15 seconds per process, 40-second budget, 16 GiB observed peak cap. Record executable hashes, raw medians/IQR/worst, peak working set and fixture parity; interactive host remains a limitation. No fresh profiler run: reuse existing attribution and measure the scoped change. Retain only if parity passes and paired timings improve beyond ordinary run variation. Report and archive.

Remaining local duplication, 2026-09-24

Created-at: 2026-09-24 01:58 +02:00. ETA: 4–6 minutes including archive. User explicitly approved both identified cleanups and a fresh duplication audit. Owner: Core. Existing task checkout fetched both remotes; no upstream drift. Remove ir/common.rs and its module declaration; migrate the two integration-test consumers to SignedRefR::target/sign. Search found no external use in the framework/Nagare source trees. Public removal is intentional; no compatibility facade, replacement module, dependency change or extra checkout.

Discovery: TransactionAssembler already owns abandon and the seven staging-limit guards. Add one private staging_limit_exceeded method there; preserve guard order, counter updates, all diagnostic strings, committed version and replay position. Strengthen existing byte-limit, chunk-limit and superseded-drain tests to verify cleanup and preserved committed/sequence state. Run the assembler unit test module, the two migrated integration-test modules, library Clippy and fmt. No benchmark or profiler is requested for this cleanup.

Compare the same token detector and scope before/after: entry 462 marked runs, 9588 marked lines, 185 scanned files. Separate source and test contributions; inspect the remaining largest Core matches before calling any a duplicate pipeline. Existing candidate inventory, report, symbol index and diagrams are updated in place; report archived to the existing Samba destination.

Shared little-endian reader, 2026-09-24

Created-at: 2026-09-24 02:12 +02:00. ETA: 5–7 minutes including archive. User explicitly requests struct/trait/generic extraction of envelope readers. Owner: Core ir_stream. Both remotes fetched; existing task checkout is current. Discovery found no shared endian reader in common or ir_stream. New private reader.rs owns LittleEndianReader over a borrowed byte slice and a LittleEndianValue trait with associated fixed byte array. Generic read performs one shared bounds-checked copy, then statically dispatches the type conversion. A compact macro supplies primitive conversions for the four actual consumers: u16, u32, u64, f32. SnapshotVersion implements the same private trait by reusing its existing public conversion. No external reader API or dependency.

Migrate envelope fields, section-table fields, typed section-body copies and snapshot counters. Remove the old rd functions and per-version closure. Keep length/resource checks before reads/allocations, preserve exact validation order and wire errors; a private reader called outside validated bounds may panic, while public decoding must reject every truncated input. No pointer casts, unsafe code, dynamic dispatch or parallel decoding implementation.

Tests: independent known bytes at unaligned offsets and exact buffer ends; integer maxima, floating-point bit patterns, invalid offsets/short input; existing envelope/section/version/codec unit tests and stream-admission integration tests. Library Clippy, fmt, symbol-index and same-scope duplication checks. No performance measurement requested. Update existing candidate inventory/report, archive evidence.

Incidence and access surfaces; assembler cleanup, 2026-09-24

Created-at: 2026-09-24 02:39 +02:00. ETA: 6–8 minutes including archive. Owner: Core. User approved incidence consolidation and further access/assembler inspection. Both remotes fetched; existing task branch has no upstream drift. Discovery finds one numeric incidence implementation already in IncidenceIndexData. Share the 14 inherent delegating methods with a private declaration template, parameterized by data lifetime: owned results borrow self, view results retain the original data lifetime. Preserve Copy/two-pointer view layout, inherent call sites, operator arithmetic/order and all public signatures. A trait-only replacement would require consumer imports; a naive generic borrow shortens view lifetimes. No new runtime abstraction, dependency,

module or compatibility API.

In access.rs share the four typed iteration/count/containment declaration families. Keep existing typed lookups and node/edge backlink checks, including the arc exception. Templates describe a repeated public surface; they do not create a second algorithm. Centralize transaction abandonment at public apply on every ResyncRequired outcome; retain diagnostic state captured before cleanup and replay bookkeeping rejection. Reset also reuses the existing abandon operation.

Extend existing incidence parity test with results surviving a short-lived view (slice, iterator, data reference and operator plan). Existing assembler test driver checks every resync preserves committed version/digest/sequence while clearing staged/draining state. Run focused incidence/access integration tests and assembler unit tests, library Clippy, formatting, symbol and duplication gates. No benchmark. Compare the same detector against 461 marked runs / 9546 marked lines / 185 files; separate declarations and test fixtures from actual pipeline duplication. Update existing inventory and report; archive to Samba.

Assembler test runner, 2026-09-24

Created-at: 2026-09-24 02:55 +02:00. ETA: 6–8 minutes: one test-only source refactor, targeted test compilation and archive. User requests runner structs for test duplication too. Owner: Core ir_stream tests. Fresh fetch of both remotes; upstream 0/0. Existing dirty checkout and all production behavior preserved.

Extend existing tests_support with AssemblerRunner owning TransactionAssembler and the existing Producer. Move the free apply helper and its real wire validation and resync invariants into the runner. Reuse that path for step, expected outcome and ordinary complete-transaction execution. Migrate unit tests, ignored taxonomy probe and model tests; keep explicit bytes for exact/conflicting replay and expose producer controls for sequence gaps, stream changes and exhaustion. No alternate validator, test-only acceptance bypass, dependency or public production API.

Retain scenario-specific assertions and reference-model independence. Remove one byte-identical duplicate replay test only after confirming the surviving test covers all its assertions. Consolidate send/apply setup where bytes are used once; keep fixtures reused by replay tests. Preserve raw message fields/order for every migrated scenario. Existing correctness tests themselves prove runner behavior.

Affected: assembler.rs test modules, existing plan bundle, candidate inventory, report and generated symbol index if needed. Source is CORE-protected; the user's current runner request authorizes this test-only change. No production sections will change. Run the assembler unit and model-test modules (taxonomy stays ignored), Clippy for the library test target, rustfmt/diff checks, symbol and same-scope duplication gates. No performance measurement. Before: 459 marked runs / 9279 marked lines, assembler 663 of which 607 in test code. A smaller source count must not be bought by weakening tests or moving identical bodies into another file. Worst cases retained: u64 sequence boundary, zero replay capacity, staged/drain limits, randomized and exhaustive message streams. Restore only this task's patch through Git if needed; no backup checkout. Append report and archive to Samba.

WGPU preparation and declaration traversal, 2026-09-24

Created-at: 2026-09-24 03:08 +02:00. ETA: 10–15 minutes: eight backend builders, two AST passes, focused correctness and archive. User explicitly approved WGPU and traversal duplicate removal. Owner: Core checkout (hymeko_wgpu backend and hymeko_core resolution). Fresh origin/github fetch; upstream 0/0; reuse checkout. Existing roadmap invariants apply: canonical IR and numerical semantics unchanged.

WGPU: OperationFactory becomes the shared borrowed context for every builder, with one OperationLimits value replacing eight per-operation limit structs. Move typed uniform allocation/upload into that existing owner. GpuTensor owns its binding-entry construction. A private OperandCheck owns the operand diagnostic label and common required-value/shape-error construction. Remove six copies of each helper and the rotor-local uniform builder. Preserve each prepare method's validation order, diagnostics, kernel/layout binding positions, workgroup policy, resource usage flags, typestate and operand lifetimes. No new preparation-time collection, extra buffer, per-encode allocation, shader or numeric change.

AST: private DeclarationFields trait in existing sym_ast.rs exposes borrowed name/bases/optional children and mutable bases/children for NodeInner/EdgeInner. An AstInterner struct owns the interner borrow; one generic declaration method visits annotation/name/bases/body in existing order, with node/edge constructors preserving None versus Some(empty). ConstValues owns one generic mutable declaration walk. Arc ref-before-annotation substitution order remains explicit; error short-circuiting and constant arithmetic stay unchanged. Remove replaced free lowering/substitution helpers. Public entry points and parser dependency remain stable; no parser checkout modification or new module.

Files: eight operations/*.rs builders, operations/mod.rs, resources.rs; Core sym_ast.rs, resolution/intern_pass.rs and const_resolve.rs; focused tests in those owners as needed; existing plan/report/candidate/symbol artifacts. These source paths are CORE-protected; the current user request authorizes them. No dependency, branch, worktree, commit or publication change. Rollback only this task's edits, preserving earlier dirty work.

Verification: Core intern/constant unit tests plus frozen identity/namespace integration fixtures; strengthen order/optional-body/error precedence checks. WGPU existing HSiKAN, entropy pooling, sparse scatter/highway/resident chain, rotor attention/spike tests plus targeted SpMM/activation preparation and dispatch cases. Inspect runners first; no profiling/benchmark opt-in. Execute GPU tests serially and require actual adapter execution when available. Clippy for both libraries, fmt/diff/symbol and duplicate gates. Before: 448 runs / 8996 marked lines / 185 files. No performance claim; shared buffers retain existing allocation sizes and ownership. Report actual results and any adapter skips; archive to Samba.

Incidence transpose row kernel, 2026-09-24

Created-at: 2026-09-24 03:24 +02:00. ETA: 6–8 minutes, one production owner, existing correctness fixtures, report and archive. User explicitly selected the remaining index/prepared-plan calculation overlap. Core-protected source changes are authorized by that request. Fresh origin/github fetch succeeded; upstream 0/0, existing dirty checkout retained. No dependency or public API change.

Introduce a private IncidenceCoefficients trait implemented by existing IncidenceView and IncidenceOperatorPlan. The live strategy resolves the canonical arc/ref coordinate; the prepared strategy reads its retained f64 coefficient. IncidenceIndexData owns one ordered transpose-row fold, shared by direct serial, prepared serial and Rayon applies. Share serial output traversal as well; preserve its initial zeroing, partial-write behavior and untouched output tail. Rayon keeps disjoint output partitioning and initial slice validation. Reuse the live coefficient strategy during preparation. Preserve coefficient multiplication and summation order, duplicate incidences, neutral references, signed zero and nonfinite policy. The direct path propagates nonfinite values; preparation rejects the first offending node in canonical node order. No dynamic dispatch, new retained array, per-apply allocation, IR copy, matrix materialization, reordered arithmetic or automatic cache. Bx and sparse topology construction remain in their existing owners.

Affected: incidence.rs, incidence_precompute_profile.rs correctness section, existing plan bundle, candidate inventory, symbol index if changed and report. Private trait and generic methods need no consumer migration. Restore only this task's patch if necessary, retaining earlier uncommitted work; no backup checkout.

Tests: reuse index/view/prepared and independent oracle fixtures. Add a compact runner covering cancellation-sensitive row order, duplicates, isolated nodes, empty graph/arc, repeat apply, output tails, overflow/neutral nonfinite coefficients and first-error node. Check bit parity including serial/Rayon; floating exceptional results by classification where NaN payloads are unspecified. Inspect aggregate selection first and keep ignored profiles disabled. Run incidence unit tests and specific incidence correctness integration filters, Clippy, scoped fmt/diff, symbol and duplication gates. Entry: 449 runs / 8543 marked lines / 185 files. No performance experiment or speed claim: current request is consolidation. Plan compile before source edit; write report and archive to Samba at completion.

Profiling continuation, 2026-10-11 17:11 +02:00. The user explicitly requests Core performance profiling and a fastest-framework objective. Measure merged Core `b66cb4d` in the existing checkout; the source tree is clean before this documentation amendment. ETA: 15–20 minutes for the bounded initial characterization, including build, repeated cells, CPU capture and analysis. Reuse `interning_bench`, `ir_bench`, `scripts/dev/measure_peak_memory.ps1`, `source_manifest.ps1` and `criterion_report.ps1`. No new benchmark owner or dependency is needed.

Initial matrix and hypotheses. First measure the existing 50k frontend phases, full/cached semantic identity and a checked leaf update. Repeat the main cells in two separate processes. After a bounded smoke, measure the existing 500k frontend's cold compile and structural hash, keeping its distinct source identity. Cover true higher-arity structure with the existing IR hash cells (1k and 10k nodes, arity 4 and 32). The competing explanations are parsing/lowering, canonical encoding, and immutable-model copying/validation; attribute each using symbolized stacks before selecting an optimization. Phase timings are not additive.

Measurement controls. Criterion release with debug information, fixed fixture hashes, raw samples, median/IQR/worst batch mean, and two repetitions. Samply records separate 10-second CPU windows for cold compilation and semantic digest/update paths; profiler times are not benchmark times. The existing Windows monitor records peak working set and observes the 16 GiB cap. Keep individual benchmark invocations below 180 seconds; log real progress, executable hashes, commands, CPU/power context and host interference. OS profiler consent may require user interaction; report it if capture is blocked. No services or other workloads are stopped. Save a timestamped report and numerical/profile/plot artifacts under `reports/2026-10-11-core-profile/`; archive with the shared report tool.

Acceptance and limits. Existing Core correctness gates are green on this revision (178 library, 152 IR verification and 7 doctests; Clippy clean). Fixture parity assertions must pass in the benchmark too. This phase changes no production code, hashes, public API or wire encoding. A profile justifies a specific subsequent optimization; it does not establish a world ranking. Such a ranking needs equivalent semantics, workloads and a separately agreed competitor matrix. RL, renderer migration and unrelated numerical/GPU campaigns stay excluded.

Optimization continuation, 2026-10-11 18:13 +02:00. The user explicitly asks to proceed with optimization and to deliver PDF reports. The completed CPU profile identifies symbol visitation and copying as update costs. Core owns both changes: `module_store/execution/value_updates.rs` lazily copies the base interner only when an update interns new text; `resolution/symbols.rs` keeps its one typed walk but marks path/alias keys without rebuilding their hash maps. Full hierarchy and model validation stay enabled. No representation, public API, hash byte, error contract, snapshot isolation or symbol-lifetime change.

Extend the existing `benches/interning_bench.rs` with an explicit digest fixture size and checked text updates. Preserve the original production binary/PDB pair after building this same harness, then change production. Reuse the benchmark's existing parity assertions; add a map-stability regression to symbol-compaction tests and run it against the old implementation first. Existing mixed/nested-value, text lifetime, invalid-input, old-reader and full/incremental parity tests remain discriminating gates. Run the 1k and 50k update workloads in paired ABBA order, with raw samples and the same Windows peak-working-set monitor. Measure full/cached digest controls at 1k. No new benchmark owner or dependency. Preserve the 16 GiB / 180-second per-process budgets.

Expected implementation, tests and measurements: 20–30 minutes. The copy-elision must retain the original symbol IDs until the existing compactor rewrites them; text edits must still own their extended interner. Marking must validate every symbol while leaving map keys and allocations intact; dense rewriting still rebuilds maps. Reject a regression exceeding 10 percent and investigate its cause. Record the result in a PDF beside the existing profiling artifacts, not a new Markdown report. Rollback only the named optimization/test/benchmark hunks, preserving prior task work. The existing checkout is reused on `perf/checked-value-update-symbols`; no new worktree. No push or merge is performed without the requested notification.

Measured continuation, 2026-10-11 18:31 +02:00. The first ABBA sequence shows numeric-update gains but no consistent text-update gain. The same symbol visitor allocates a worklist even for a numeric or DeclId reference scalar, neither of which contains a SymId. Return immediately for these two scalar roots; string/list traversal and all validation remain unchanged. Preserve the first measured candidate and use separately named final ABBA samples, against the same original binary and unchanged benchmark harness.

Second optimization, 2026-10-11 19:05 +02:00. The user authorizes the next measured Core optimization. The previous candidate's symbolized update profile attributes 30.5 percent of selected stacks to hierarchy coherence validation: copied path vectors, sorting and a second lookup table. Core's existing `ir/coherence.rs` owns this validator; `PathKey` already supports borrowed symbol-slice lookup. Borrow the existing paths and query the existing Index instead of allocating per-path vectors and rebuilding a HashMap. Keep deterministic declaration/path ordering with an unstable sort: Index keys are unique, so ties cannot occur. All checks and report limits remain active. Typed SymId paths also avoid the previous lossy u32 conversion; add a regression for distinct 64-bit symbols with identical low 32 bits, plus exact ordered/capped findings across reversed index insertion and nearest-prefix/empty-path cases. This intentionally corrects false prefix matches for malformed wide symbol IDs; ordinary graph findings, public API and hash/serialization algorithms are unchanged.

Extend the existing coherence integration suite. Reuse the preserved first-candidate EXE/PDB as baseline and the unchanged interning benchmark. Run bounded 1k/50k smoke, then numeric/text update ABBA comparisons and the 1k digest controls; 16 GiB cap, under 180 seconds per process. Capture the optimized CPU profile and preserve source manifests. Run Core library/unified-IR correctness, Clippy and downstream checks. Report this second step and cumulative gains separately in PDF. Estimated duration: 15–25 minutes including build, paired measurements and reporting. Reuse this dirty task checkout and branch; preserve prior changes. Rollback only this continuation's coherence/test hunks. No commit, push or merge is requested by this continuation.

Third continuation, 2026-10-11 19:26 +02:00. The user requested commit/push and continued optimization. The previous two steps are published as `8a7d7f5` on `perf/checked-value-update-symbol`. The latest profile attributes 37.4 percent to deep IR cloning, principally owned incidence and weight vectors. Sharing those public Vec fields requires a consumer API migration, which is not

silently bundled into this step. The next compatible copy reduction targets Interner construction (23.4 percent of update samples).

Core's existing `resolution/interner.rs` owns a private reference-counted table and immutable strings. Existing public methods remain unchanged; private fork and retained-symbol operations share unchanged data. Insertion detaches a shared table. `resolution/symbols.rs` retains the one typed symbol walk and uses the Interner-owned dense ID mapping. `execution/value_updates.rs` forks the symbol table lazily when new text is introduced. No old table is retained by a compacted table with dead strings. This also removes the fabricated static string references from the old interner. Tests cover detached mutation, dense ID mapping, dropped dead-string ownership, retained readers and existing digest/CBOR parity. No dependency, public signature or canonical encoding change.

Reuse the preserved `after-coherence/` EXE/PDB as baseline and the unchanged Criterion harness. Measure 1k/50k numeric and text updates, 1k digest controls, and 50k intern/cold-compile controls in ABBA order after bounded smoke. All cells retain the 16 GiB and 180-second process bounds. Reject unexplained regressions over 10 percent, including the compiler and memory controls. Capture a symbolized CPU profile, run owning and downstream correctness, and deliver a separate PDF. Estimated work: 20–30 minutes, including construction-regression measurements. Rollback only this continuation's Interner/SymbolCompaction/update hunks; keep the published first two steps and their evidence. The same checkout is reused.

Shared IR continuation, 2026-10-11 20:47 +02:00. The user explicitly authorizes Core shared storage plus Graph and Transformation adaptation; other consumers follow only after Core. Preserve the pending shared Interner changes on the current branch (HEAD `8a7d7f5`); fresh origin fetch succeeded, branch upstream is equal, main is one ancestor commit behind. Graph `ed7d0cf` and Transformation `90d64a8` match their freshly fetched feature upstreams. Transformation main diverges by 7/1 (the other commit is a material-lobes merge); reuse its existing feature integration base without discarding that branch's work or silently merging main. Existing TeX byproducts in both consumer checkouts remain untouched.

Core owns `ir/table.rs`: one contiguous `IrTable<T>` over shared Vec storage, copied on mutable access. Existing read accessors and ordinary indexing, iteration, append and slice operations stay available. Explicit Vec assignments or typed clone results may require conversion. Ir clone shares all nine tables; a leaf edit copies the declaration table only. This is table-granular, not paged or constant-time leaf editing. Meta stays unchanged. Serialization delegates to the same vector sequence; existing manual wire admission remains authoritative.

`resolution/symbols.rs` uses one macro-defined field walk instantiated with shared or mutable borrows; marking cannot detach any Ir table. Dense rewriting may detach tables when symbol IDs actually move. All validations stay active. Other Core edits: `ir/mod.rs`, `ir/ir.rs`, direct Vec assignments in `module_store/module_store.rs`, existing update/symbol tests and any compiler- identified table-boundary callers. Update the generated symbol index. Graph's explicit owned arc clone uses an explicit deep copy; adapt further proven consumers only after Core gates pass. Temporary local patches are integration aids, not published dependency pins. Do not publish an uncommitted provider revision.

Before creation: table preflight found StringTable; rejected because it is an immutable string/Python snapshot, not an editable generic IR record table. The generational-arena artifact is a retired representation experiment; no code is copied from it. Reuse existing plans, benchmarks, fixtures and tests. Add tests for mutable access/iteration/owned iteration isolation, sequence wire parity, retained old snapshots and unchanged-table sharing, including read-only symbol marking and dense remapping. Run Core correctness/Clippy first, then Graph and Transformation workspaces with one local Core identity. Reuse after-sharing EXE/PDB baseline; smoke 1k/50k, ABBA update and cold-compile/intern controls, existing IR hash/traversal controls, separate CPU profile, 16 GiB/180-second process bounds. Preserve all raw evidence and deliver PDF. Estimate 45–75 minutes: storage/walk implementation, ownership tests, consumer compilation and paired

measurements. Rollback only these storage/migration hunks, retaining the prior Interner step.

Construction-control correction, 2026-10-11 21:00 +02:00. The first Arc-backed table candidate keeps cold compilation within 0.2 percent but regresses direct IR construction by 25 percent; it is not accepted. Preserve its binary and results. Keep a uniquely owned Vec during construction/mutation, and convert it without copying records to shared storage at CompiledProgram and CompiledModel publication boundaries. A first write detaches shared storage once; later writes use the owned Vec directly. Standalone mutable tables clone deeply; published snapshots share. Both preserve independent Clone semantics. No unsafe shortcut or reference-count bypass. Recheck the same construction/read controls, then run the update matrix. The stored earlier `ir_bench` baseline is valid for these controls because the preceding Interner/coherence changes do not enter their timed construction, raw state fold or accessor traversal; input and benchmark source are unchanged. Full digest/compiler/update controls use after-sharing.

CPU closure, 2026-10-11 23:33 +02:00. Windows consent was granted and both delayed captures completed. The rejected Arc-table construction candidate contributes 6,273 selected construction samples; 26.7 percent include Arc mutable-access handling. The final shared-IR update capture contains 9,143 samples, 4,850 in the actual value-update function. All 813 sampled Rust addresses resolve against the preserved final EXE/PDB. The final binary and all 191 manifest files still match the measured candidate; no production code or timing matrix is changed in this closure.

Compare actual copy stacks, not only a function name whose work moved: the union of whole-Ir clone and declaration/node/edge/arc Vec clone frames falls from 52.2 to 16.9 percent of selected update samples. Topology-table copying has no samples in the final numeric workload; declaration-table detachment remains. Inclusive shares overlap and are not additive phase times. Keep the measured accessor traversal tradeoff and the independently reproduced pre-existing Transformation fingerprint failure visible. Update the existing PDF/JSON and archive the new profiles, resolved symbols and figures. No new commit or push is authorized here.

Algorithms and storage, 2026-10-12 01:09 +02:00. The user requests algorithm and data-structure selection followed by implementation, including heap storage. Fresh Core origin fetch succeeded; HEAD and feature upstream remain at 8a7d7f5, one commit ahead of main. Reuse the checkout and retain the pending shared Interner and IR work. Core owns canonical skeleton preparation and update bookkeeping; Graph owns traversal queues/workspaces; Tensor already owns numerical buffer arenas. No sibling source changes, RL work or publication.

Implement stable counting/prefix-sum packing in the existing SkeletonForest: replace one child Vec per declaration with a flat child-ID buffer and N+1 offsets. Reverse stable scatter may reuse cumulative row ends as cursors, leaving row starts without an extra scratch array. Preparation is O(N+E), row lookup O(1), row iteration O(degree), and exactly two child-storage buffers replace the outer vector plus nonempty row buffers. Keep the existing priority heap, labels, bottom-up fold, hash bytes and topology preconditions. No public signature change. The generic numerical CSR owner is not an IR hierarchy owner; introduce no new generic CSR abstraction. Searches found only this private child-list owner.

Affected files: this plan bundle, CORE.YAML, SYMBOLS.md if generated output changes, and `hymeko_core/src/ir/canonical_hash.rs`, including private correctness tests. Add independent row-oracle checks for empty, chain, star, mixed named/anonymous, out-of-range-parent and cyclic inputs, and exhaustive small parent maps. Run focused tests, existing frozen-hash/corpus/semantic identity checks and owning library checks; run formatting, Clippy and symbol/duplication gates. Worst inputs include a chain with a heap buffer for every original nonleaf, a wide star and mostly anonymous declarations. Offsets cover the full declaration-ID domain.

Deliver a Hungarian PDF catalogue with primary references, applicability, worst-case versus amortized bounds, allocation/reclamation costs and implementation order. Investigate delta propagation, inverse symbol-use indices, sparse sets, indexed heaps, bounded pools, TLSF and snapshot reclamation. Deadline/hardware/OS, maximum graph size and retained-reader bounds are still unspecified; ask for them without blocking this deadline-independent storage work. No new timing campaign; report analytical storage bounds separately from measured performance. ETA about 25 minutes: implementation/tests 10, sourced catalogue and PDF verification 15. No numerical latency improvement or hard-real-time guarantee is asserted. The existing 16 GiB workspace resource cap remains. Roll back only this continuation's named hunks, preserving earlier changes. Archive the final PDF and verification evidence through the existing report-sync tool.

Packed-child correctness closure, 2026-10-12 01:16 +02:00. Implemented in the existing private owner. Two new tests compare against a direct parent-scan oracle: all 3,529 kind/parent combinations for zero through three declarations, plus 257-vertex chain, star and branching cases. Core library: 187 passed, 2 ignored. Unified IR: 181 passed, 7 ignored, including frozen hashes, Pruefer cycle rejection and corpus CBOR parity. Strict Clippy and changed-file formatting pass; symbol index is current. Duplication remains 448 clones / 8,529 lines, below the baseline 484 / 10,411. No benchmark or new speedup claim. Preflight's filename candidate `hymeko_wgpu/tests/skeleton.rs` is a GPU backend test suite, not a named-declaration hierarchy representation; rejected. No new production type or file was introduced. The implementation and tests are covered by source manifest `322c53db7343f2572b5c7f5cb9a942fe710991bdee16e0ee843f8bc54e70838e`. The algorithm/heap/allocation catalogue is delivered as `reports/2026-10-12-algorithms-storage.pdf`; selected future candidates are explicitly distinguished from this implemented storage change.

Approved catalogue continuation, 2026-10-12 01:31 +02:00. The user authorizes every catalogue candidate. Implement in owner order, Core before consumers; retain conditional applicability (no arbitrary global allocator replacement or invented hard deadline). Fresh origin fetch: feature HEAD/upstream 8a7d7f5 equal, one ahead of main, existing checkout retained. No new publication instruction is inferred. Preserve pending shared storage and packed-child work.

First deliver shared CompiledProgram Index/import ownership through Arc, with deep detach only when symbols are renumbered. CompiledProgram public field types therefore change; borrowed access coerces as before, owned construction needs conversion. Core CBOR wire payload keeps its existing owned Index/Vec types. The workspace search found one external constructor in hymeko_web's WASM document owner; adapt only after Core gates, with a fresh fetch and local rules. Core direct consumers include pretty-print and namespace-fault tests.

Cache exact model-local symbol-use counts lazily in CompiledModel. Generalize the one macro-defined field walk to read visitors; use it for full marking, counting and value deltas. Collect signed symbol occurrences only from changed old/new leaf values, sort by symbol ID and combine each run before adjusting counts. Equal multisets keep the shared count vector. With no unused symbol, reuse Interner and metadata without a full mark/retention scan. If a symbol dies, reuse the existing dense retention/rewrite path and compact the count vector in the same order. Retain all coherence/model validation and final digest folding. Numeric updates avoid count-vector copying after lazy initialization. Text updates may still copy O(S) counts/interner data and globally remap IDs: do not claim O(k) for the whole operation. Sorting a delta of R occurrences costs O(R log R).

Touched: CORE.YAML; this plan bundle; SYMBOLS.md generated index; `resolution/symbols.rs`; `module_store/module_store.rs`; `module_store/execution.rs`; `module_store/execution/value_updates.rs`; `writers/cbor_writer/tests.rs`; `util/pretty_print.rs`; `tests/unified_ir/verification/namespace_faults.rs`; compiler-identified direct Core callers. Reuse existing tests and add multiset/count parity against the independent full mark/compaction path;

retained readers, no-op, scalar, nested lists, shared text, dead suffix, dense remap, invalid symbols, and imports/aliases. Check pointer sharing and count state without timing assertions. Run targeted Core library/IR/Clippy gates, then affected consumers. PDF evidence and archive. ETA 35–50 minutes for this first dependency pair: implementation 15, correctness 10, integration/report 10–25. No profiling matrix in this step. Existing 16 GiB resource cap applies. Roll back only named continuation hunks; preserve earlier work.

Parent-chain algorithm, 2026-10-12 01:40 +02:00. The first pair passes 191 library and 181 unified-IR tests plus Clippy. Continue within the approved catalogue: the existing parent-acyclicity validator repeats each ancestor chain, O(N times depth), or O(N times max-walk) on cycles. Classify each functional-graph vertex once with unseen/active/completed state and unwind an explicit path stack. Completed outcomes retain distance to root or invalid parent, or cycle reachability. Emit diagnostics afterward in the original declaration order and with the same capped collector. O(N) parent classification, O(N) scratch; other validator work is unchanged. This is not validation caching: every current input is still checked. No identity-equality shortcut.

Keep legacy diagnostic walked values for finite limits: overlong/rootless walks report max-walk plus one, invalid parents report the earlier invalid distance. At usize::MAX, saturate that diagnostic bound and terminate cycles instead of overflowing or looping. Modify only `ir/coherence.rs` plus this plan/manifest. An independent bounded naive-walk oracle checks every parent map up to four vertices at limits 0, 1, N and 2N; include cycles, out-of-range parents and NONE. Test usize::MAX cycle termination and existing full-validator faults. Add about 10 minutes implementation/correctness. Retain all previously stated gates.

Algorithm implementation closure, 2026-10-12 02:05 +02:00. Core: 193 library, 181 unified-IR and 7 doctests passed; 9 ignored; strict Clippy passes. All 1,441 parent maps through four vertices match the bounded naive diagnostic oracle at four walk limits and three report caps. Delta tests cover full counts/compaction parity, repeated symbols, aliases/facets/imports, suffix removal, remapping, retained readers and count errors. Source manifest (191 files): `e516c5e60c5d2f37f2b44d3f`

Graph advanced externally to f7ed8c2 during integration; retain it and its published 605bc8c arena-test fix. The separately planned indexed frontier passes 644 Graph tests (30 ignored), including the allocation and path oracles, plus strict Clippy. Transformation current feature 0c1a2c5 contains the external emitter fingerprint correction: 330 selected tests pass (5 ignored, workflow timing test explicitly filtered), and all workspace test targets compile. An earlier suite invocation also ran its non-ignored workflow timing test; no performance claim is drawn from that incidental measurement. Generated SDF blank lines restored only after checking exact provenance/content. Web 4bfe66a needs three ownership conversions; 42 native tests and the wasm32 target check pass. All temporary lockfile changes were restored after hash checks. Single local provider identities were verified; these remain local integration builds, not publication pins.

Core duplication remains 448/8,529 below 484/10,411. Graph's stock gate fails against 547/12,629; current HEAD and candidate both measure 577/13,157 using the same provider detector. The heap adds zero clones; do not raise the baseline or describe that gate as green. Current symbol indices and formatting pass. PDF and evidence: `reports/2026-10-12-incremental-bookkeeping.pdf`, with Graph's copy and source evidence under its indexed-frontier report. Other catalogue candidates remain authorized; deadline/platform/capacity requirements were requested for the memory/publication work. No commit, push or merge by this continuation. Existing uncommitted work and user TeX outputs retained.

Commit and requirements clarification, 2026-10-12 02:14 +02:00. The user requests committing the completed changes. Prepare local commits in provider order (Core, Graph, Web); no new push or merge is requested. Existing test/source manifests identify the reviewed candidate. The user clarifies that hardware, deadlines, graph/delta limits and retained-reader limits are not fixed at this early framework stage. Continue with platform-independent APIs and configurable

capacities, refining concrete limits with later use. These missing fixed values are therefore not a reason to stop the authorized algorithm/storage work or request the same approval again. Do not invent hard-real-time guarantees.